probability $\widetilde{\omega}_j$. Because the external deadline is $W \sim \text{Cat}(\boldsymbol{\omega})$, we can write $\widetilde{\omega}_j$ as

$$\widetilde{\omega}_j \equiv \Pr(W = i \mid W \geq i) = \frac{\omega_i}{\sum_{j=i}^{W^{max}} \omega_j} \tag{1}$$

For a given token, the step at which the model must generate the result —self- or world-induced—is a random variable, S, with probability given by

$$\Pr(S = i|\boldsymbol{d}, \boldsymbol{\omega}) = \left(\omega_i + (1 - d_i) \sum_{j=i+1}^{W^{max}} \omega_j \right) \prod_{j=1}^{i-1} d_j \tag{2}$$

and $d_{W^{max}} \equiv 0$ for the final step. Equation (2) follows from Figure 3; see Appendix A for the derivation.

The expected discounted accuracy obtained by the model in the CYB decision task is $\mathbb{E}_{i \sim \Pr(S|\boldsymbol{d},\boldsymbol{\omega})} [\gamma_i t_i]$, which can be turned into a negative log likelihood loss:

$$\ell_{\text{CYB}}(\boldsymbol{t}, \boldsymbol{d};\ \boldsymbol{\omega}, \boldsymbol{\gamma}) = -\log \mathbb{E}_{i \sim \Pr(S|\boldsymbol{d},\boldsymbol{\omega})} [\gamma_i t_i] \,. \tag{3}$$

The loss depends on model outputs $\boldsymbol{t}$ and $\boldsymbol{d}$ and on hyperparameters $\boldsymbol{\omega}$, the distribution over number of steps allotted for a token, and $\boldsymbol{\gamma}$, the vector of step-wise discount factors. The loss is equivalent to the standard cross-entropy loss with the additional twist that responses can be read out at any step $i \in \{1, ..., W^{max}\}$ and accuracy at response step i is discounted by γ_i. The expectation in Equation 3 is computable by summing over the W^{max} steps to compute a mean of $\gamma_i t_i$ weighted by the probability in Equation 2.

In earlier work with pause tokens (Goyal et al., 2024), a cross-entropy loss was used at the final pause step, which is a specific instantiation of our loss with <DON'T KNOW> fixed (i.e., $d_i = 1$ for $i \in \{1, ..., W^{max} - 1\}$), reward undiscounted (i.e., $\gamma_i = 1$ for $i \in \{1, ..., W^{max}\}$), and no early stopping (i.e., $\omega_i = 0$ for $i < W^{max}$).

When an LLM is augmented to output <DON'T KNOW>, Equation 3 can be used to train both the <DON'T KNOW> probabilities, $\{d_i\}$, and the next-token distribution, via $\{t_i\}$. However, this loss won't necessarily encourage the model to use compute to good end. For example, if reward is undiscounted (i.e., $\gamma_i = 1$) and delays are always granted ($\omega_i = 0$ for $i < W^{max}$), the same reward is obtained whether a given response is produced on step 1 or on step W^{max}. The model has no incentive to adhere to a computational budget and calibrate its use of <DON'T KNOW> to situations where additional time would improve the response.

We thus propose three instantiations of the CYB loss that promote sensitivity of pause requests to model performance, aiming to obtain a *speed-accuracy trade off*, i.e., performance that improves monotonically with additional compute. First, CYB-AP tries to ensure *anytime prediction* accuracy matched to a world stop-time distribution. Second, CYB-VA is a *variational approach* that allows the model to internalize a stop-time distribution as a prior. Third, CYB-DP imposes a *distributional penalty* on the number of computation steps. We summarize the three approaches in Table 2.

Table 1: Notation Used in Article

Term	**Meaning**
W^{max}	maximum number of steps granted by world for any token
W	r.v. denoting # steps granted by world for a specific token, $1 \leq W \leq W^{max}$
S	r.v. denoting # steps at which model generates result (either by world or model <DON'T KNOW>), $1 \leq S \leq W^{max}$
ω_i	prob. that world terminates processing at step i, $\Pr(W = i)$
$\widetilde{\omega}_i$	prob. that world terminates processing at i given no prior termination, $\Pr(W = i \mid W \geq i)$
d_i	prob. of model abstention (<DON'T KNOW>) at step i
s_i	prob. of outputting result (self- or world-induced) at step i; shorthand for $\Pr(S = i)$
t_i	prob. that the model selects ground-truth token at step i
γ_i	discount factor on accuracy at step i, $0 \leq \gamma_i \leq 1$
ρ_i	[CYB-VA, CYB-DP] prior prob. of self-induced termination at step i
α	[CYB-DP] penalty coefficient on prior-distribution mismatch

Table 2: Three Catch-Your-Breath Variations

Variation	**World-induced stop-time distribution ($\boldsymbol{\omega}$)**	**Discount factor (γ)**	**Other hyperparameters**
Anytime Prediction (CYB-AP)	hyperparameter	hyperparameter	W^{max}
Variational Approach (CYB-VA)	$\omega_i = \begin{cases} 0 & \text{if } i < W^{max} \\ 1 & \text{otherwise} \end{cases}$	$\gamma_i = 1$	W^{max}, $\boldsymbol{\rho}$
Distributional Penalty (CYB-DP)			W^{max}, $\boldsymbol{\rho}$, α

2.1 Anytime Prediction (CYB-AP)

Anytime prediction requires that a model be prepared with a response at every computation step. To optimize for anytime prediction, the loss must be sensitive to the model output at each step. Equation 3 has this property given a stop-time distribution, $\boldsymbol{\omega}$, with nonzero probability at each step, i.e., $\omega_i > 0$ for all i. To minimize this loss, the model should learn to request additional compute only when discounted accuracy at step $i+1$ is no lower than on step i: $\gamma_{i+1} t_{i+1} \geq \gamma_i t_i$. With a monotonicity constraint on the discount factor, $\gamma_{i+1} \leq \gamma_i$, the model should learn to calibrate its <DON'T KNOW> probability to its discounted accuracy. We can impose a linear cost per step with $\gamma_i = 1 - (i-1)(1-\gamma_0)$ or exponential with $\gamma_i = \gamma_0^{i-1}$ for a given value of γ_0. We refer to the CYB loss with these constraints on $\boldsymbol{\omega}$ and $\boldsymbol{\gamma}$ as CYB-AP, for anytime prediction.

By incorporating the externally imposed stop-time distribution $\boldsymbol{\omega}$, CYB-AP has an upper limit on the expected number of compute steps, independent of model behavior: the model can reduce the number of compute steps but it cannot increase them. In the two approaches we describe next, the externally imposed upper limit is eliminated by setting $\omega_i = 0$ (equivalently, $\widetilde{\omega} = 0$) for $i < W^{max}$ and is replaced by soft constraints that encourage the model to adhere to a desired stop-time distribution on its own. We distinguish this distribution to be internalized from $\boldsymbol{\omega}$ using the notation $\boldsymbol{\rho}$.

2.2 Variational Approach (CYB-VA)

We wish to train the model to produce a desired distribution $\boldsymbol{\rho}$ and to minimize the loss with respect to this distribution, i.e.,

$$\ell_{\text{CYB}\rho} = -\log \mathbb{E}_{i \sim \text{Cat}(\boldsymbol{\rho})} \left[\gamma_i t_i\right] .$$

However, this loss does not depend on $\boldsymbol{d}$ and will thus fail to train the <DON'T KNOW> output, let alone train it to be consistent with $\boldsymbol{\rho}$. To achieve this goal, we take a variational approach and train the model using the ELBO. Treating the model's likelihood at step i as the penalized value $\Pr[\text{target}|S=i] = \gamma_i t_i$, the ELBO is the right hand side of the inequality:

$$\log \mathbb{E}_{i \sim \text{Cat}(\boldsymbol{\rho})} \left[\gamma_i t_i\right] \geq \mathbb{E}_{i \sim \text{Cat}(\boldsymbol{s})} \log \left[\gamma_i t_i\right] - D_{\text{KL}}(\boldsymbol{s} \,||\, \boldsymbol{\rho})$$

where we assume as before that externally induced stops are eliminated. The <DON'T KNOW> distribution, $\boldsymbol{d}$, is incorporated via the stopping time distribution $\boldsymbol{s} \equiv \Pr(S|\boldsymbol{d})$ (Equation 2). We then define a negative ELBO loss,

$$\ell_{\text{CYB-VA}} = -\mathbb{E}_{i \sim \text{Cat}(\boldsymbol{s})} \log \left[\gamma_i t_i\right] + D_{\text{KL}}(\boldsymbol{s} \,||\, \boldsymbol{\rho}) .$$

As an upper bound on our desired loss function, $\ell_{\text{CYB}\rho}$, minimizing $\ell_{\text{CYB-VA}}$ trains the model's predictions ($\boldsymbol{t}$) by optimizing the expected discounted likelihood under $\boldsymbol{s}$. It also trains the model's stopping time distribution ($\boldsymbol{s}$) toward the posterior, following the optimization view of Bayesian inference (e.g., Zellner, 1988). We can determine the equilibrium self-stop distribution by taking the gradient of the variational loss,

$$\nabla_{\boldsymbol{s}} \, \ell_{\text{CYB-VA}} = -\log \boldsymbol{\gamma} - \log \boldsymbol{t} - \log \boldsymbol{\rho} + \log \boldsymbol{s} + \mathbf{1} .$$

Introducing a Lagrange multiplier for the constraint $\sum_i s_i = 1$ yields the optimality condition $\log \boldsymbol{s} = \log \boldsymbol{\gamma} + \log \boldsymbol{t} + \log \boldsymbol{\rho} - (\lambda+1)\mathbf{1}$, resulting in $s_i \propto \rho_i \gamma_i t_i$. The model will thus learn to stop at times when both the stopping prior and its discounted accuracy are high. This analysis also indicates that the prior, $\boldsymbol{\rho}$, and the discount factors, $\boldsymbol{\gamma}$, are redundant in CYB-VA. We thus assume $\gamma_i = 1$ in all experiments with CYB-VA. CYB-VA thus has hyperparameters $\boldsymbol{\rho}$, in addition to specifying W^{max}, the upper limit on the number of steps.

2.3 Distributional Penalty (CYB-DP)

A final approach is to optimize for our original loss, ℓ_{CYB} (Equation 3), and impose a penalty based on the deviation of the model's desired and observed stop-time distributions:

$$\ell_{\text{CYB-DP}} = \ell_{\text{CYB}} + \alpha \, D_{\text{KL}}(\boldsymbol{\rho} \,||\, \boldsymbol{s}) \,,$$

where α specifies a penalty on the distribution mismatch. Note that the direction of the KL terms is reversed for $\ell_{\text{CYB-DP}}$ versus $\ell_{\text{CYB-VA}}$, in order to strongly penalize stopping times that are outside the support of the prior. (And we wish to allow the prior to have zero values.) CYB-DP has hyperparameters $\boldsymbol{\rho}$ and α, in addition to specifying W^{max}, the upper limit on the number of steps.

3 Related research

Pause tokens. Herel and Mikolov (2023) proposed the idea of special 'thinking' input tokens and Goyal et al. (2024) independently explored 'pause' tokens. In Goyal et al., a fixed number of pauses were inserted into the text stream by the LLM controller. During pretraining, blocks of W tokens were inserted at uniformly random positions. When pauses are inserted, model outputs for the $M-1$ steps leading up to the final pause token are ignored and excluded from the usual next-token-prediction cross-entropy loss. During fine tuning, where a prefix sequence is given along with a target sequence, M' pauses were appended to the prefix, giving the model additional compute steps before its response. Goyal et al. found consistent benefits in downstream-task accuracy only when pauses were included in both pretraining and fine tuning stages. Although this approach appears to be an alternative to chain-of-thought methods, pauses may be related given that substituting meaningless filler tokens for the chain of thought still yields benefits (Pfau et al., 2024). In all of these works, long delays are introduced between a prefix and a model's response, intended to support explicit reasoning about and calculation of the response—an alternative to chain of thought. In contrast, our motivation is to allow short delays in processing any token, with the aim of facilitating integration of information in prefix tokens as well as in choosing any token in the response. Similar in spirit to our proposal, also focused on improving next-token prediction, is a method that involves generating reasoning traces for individual tokens (Dong et al., 2025). The computational cost of our technique is significantly lower.

Anytime prediction and adaptive computation. In the anytime-prediction literature, models are tasked with making a series of responses that improve over time or with additional compute. Such models exhibit a *speed-accuracy* trade off in which a coarse answer appears quickly but is refined with additional computation. In some schemes, the model is rewarded for getting the right response in as few steps as possible (e.g., Iuzzolino et al., 2021). Anytime prediction is related to adaptive computation-time for recurrent networks (Graves, 2016).

Adaptive depth models. Extending adaptive computation-time to transformers, methods have been proposed in which computation is dynamically allocated by token, effectively making the model deeper for hard tokens (Bae et al., 2025; Raposo et al., 2024). The allocation is performed by a learned router, trained on the standard loss, rather than our approach which is based on a well-formed model of computation with an integrated loss that considers speed-accuracy trade offs.

<DON'T KNOW> *outputs.* The <DON'T KNOW> output is related to classification with absention (or rejection) (Bartlett and Wegkamp, 2008; Liu et al., 2019; Hendrickx et al., 2024). Two recent articles introduce heuristic notions of "I don't know" to determine where pause tokens should be inserted. Cohen et al. (2024) include a <DON'T KNOW> output from the LLM and introduce a heuristic training objective that shifts probability to the <DON'T KNOW> output when the model predicts incorrectly. Their approach applies multiple heuristics to prevent the degenerate solution of always outputting <DON'T KNOW>. By integrating the <DON'T KNOW> option into our sequential decision task, our approach does not suffer such degenerate solutions. Kim et al. (2025) insert pause tokens when output entropy is high. This heuristic approach cannot distinguish aleatoric from epistemic uncertainty, which the CYB loss can in principle infer.